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**Yi**

(10) **Patent No.:** **US 12,389,682 B2**
(45) **Date of Patent:** **Aug. 12, 2025**

(54) **DISPLAY PANEL AND METHOD FOR FABRICATING THE SAME**


(71) Applicant: **WUHAN CHINA STAR OPTOELECTRONICS SEMICONDUCTOR DISPLAY TECHNOLOGY CO., LTD.**, Hubei (CN)

(72) Inventor: **Shijuan Yi**, Hubei (CN)

(73) Assignee: **WUHAN CHINA STAR OPTOELECTRONICS SEMICONDUCTOR DISPLAY TECHNOLOGY CO., LTD.**, Hubei (CN)


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(51) **Int. Cl.**
***H10D 86/60*** (2025.01)
***H01L 25/075*** (2006.01)
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CPC ......... ***H10D 86/60*** (2025.01); ***H01L 25/0753*** (2013.01); ***H01L 25/167*** (2013.01); ***H10D 86/443*** (2025.01); ***H10H 20/857*** (2025.01)

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*Primary Examiner* — Matthew C Landau
*Assistant Examiner* — Nora T. Nix


(57) **ABSTRACT**

A display panel and a method for fabricating the same are disclosed. The display panel includes a base substrate, a micro-light emitting device layer and a thin film transistor array layer. The micro-light emitting device layer is disposed on one side of the base substrate, a light output side faces to the base substrate, and the thin film transistor array layer is formed on the side of the micro-light emitting device layer facing away from the light output side. LED chips do not need to be bound and connected to the thin film transistor array layer through a conductive adhesive bonding process
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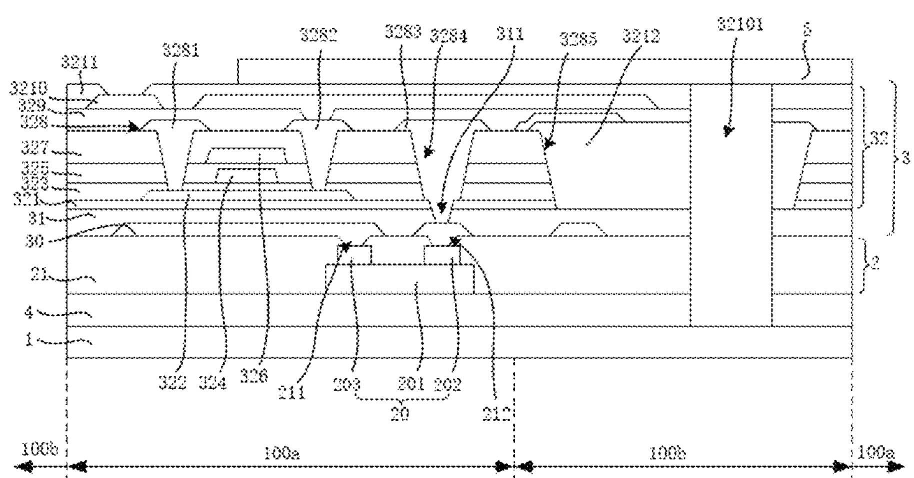

or a metal bonding process during mass transfer, so that the stability and yield are improved.

**10 Claims, 8 Drawing Sheets**

(51) **Int. Cl.**

| | |
|---|---|
| ***H01L 25/16*** | (2023.01) |
| ***H10D 86/40*** | (2025.01) |
| ***H10H 20/857*** | (2025.01) |

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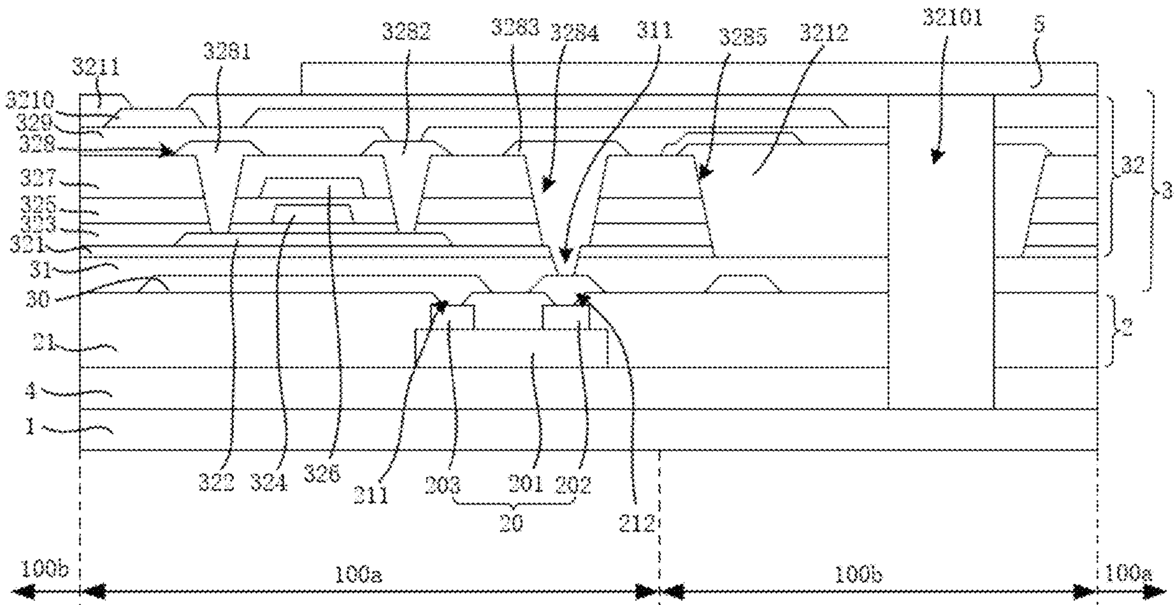
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FIG. 1

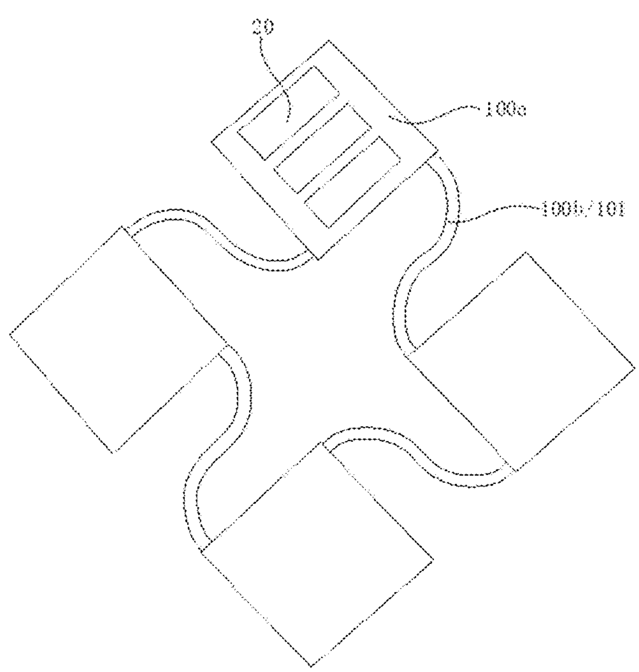
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FIG. 2

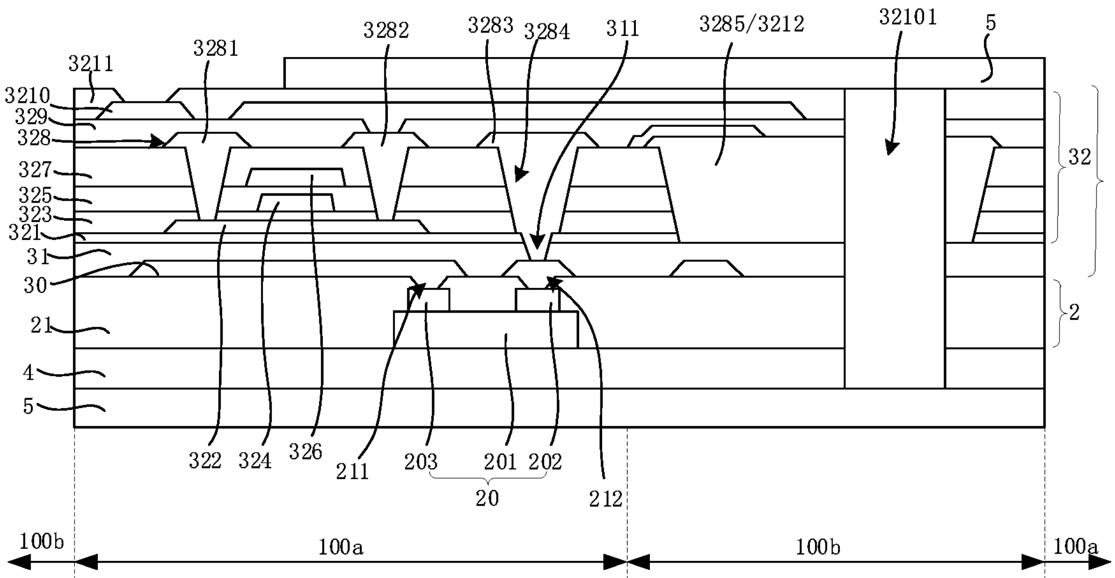
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FIG. 3

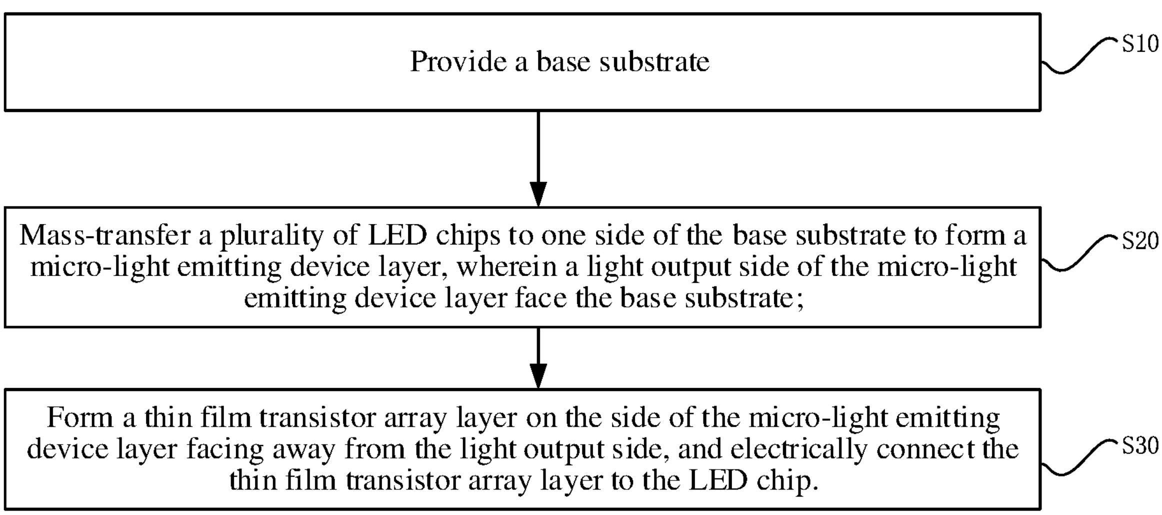
Provide a base substrate
S10
Mass-transfer a plurality of LED chips to one side of the base substrate to form a micro-light emitting device layer, wherein a light output side of the micro-light emitting device layer face the base substrate;
S20
Form a thin film transistor array layer on the side of the micro-light emitting device layer facing away from the light output side, and electrically connect the thin film transistor array layer to the LED chip.
S30

FIG. 4

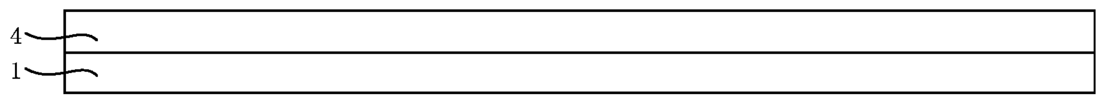


FIG. 5A

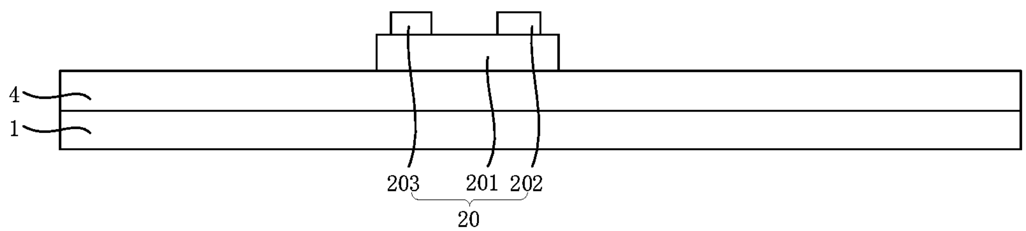


FIG. 5B

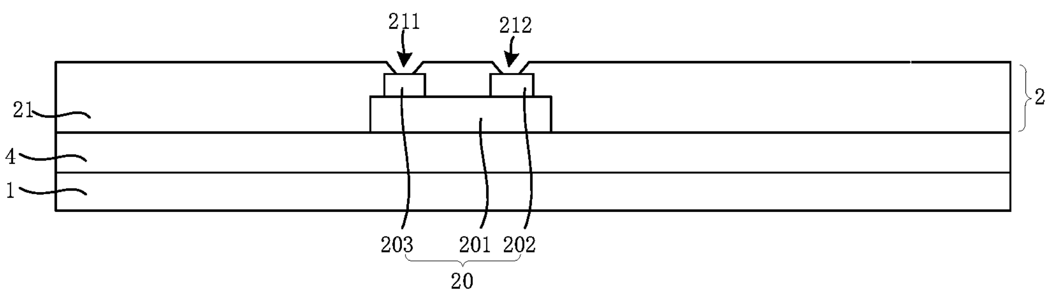


FIG. 5C

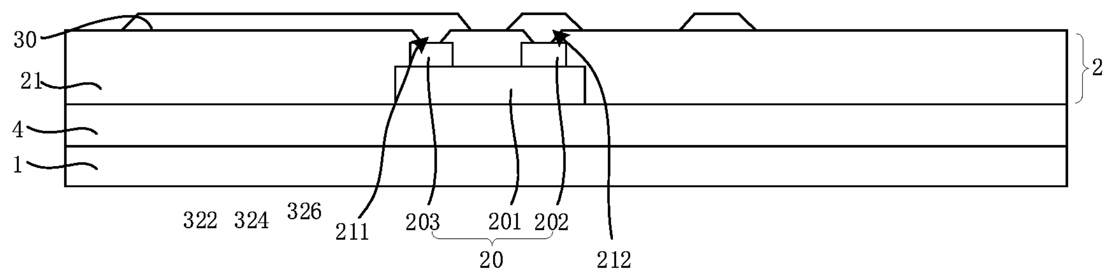
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FIG. 5D

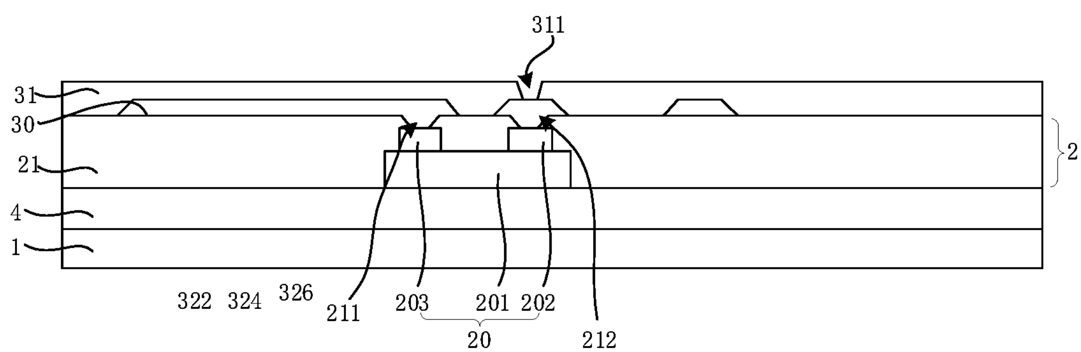
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FIG. 5E

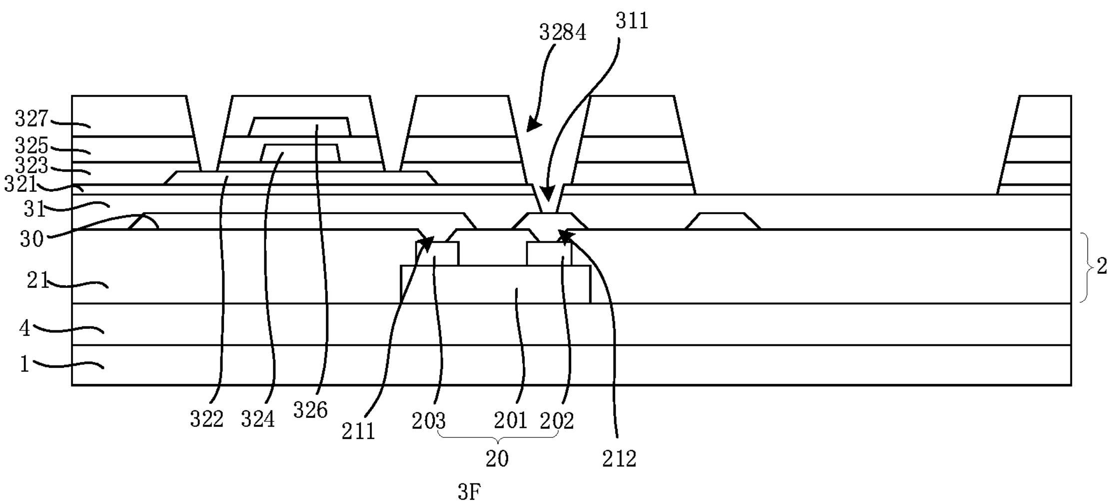
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FIG. 5F

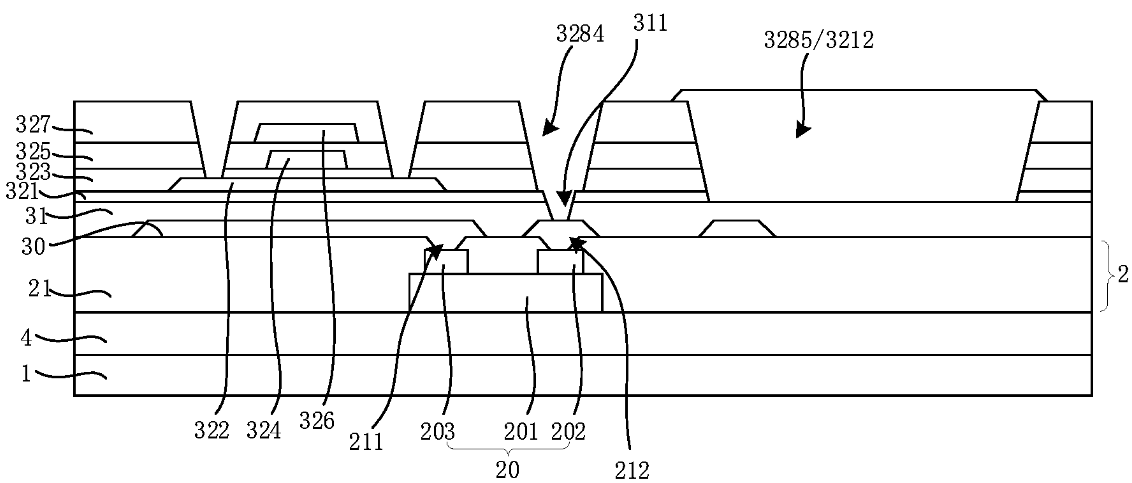
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FIG. 5G

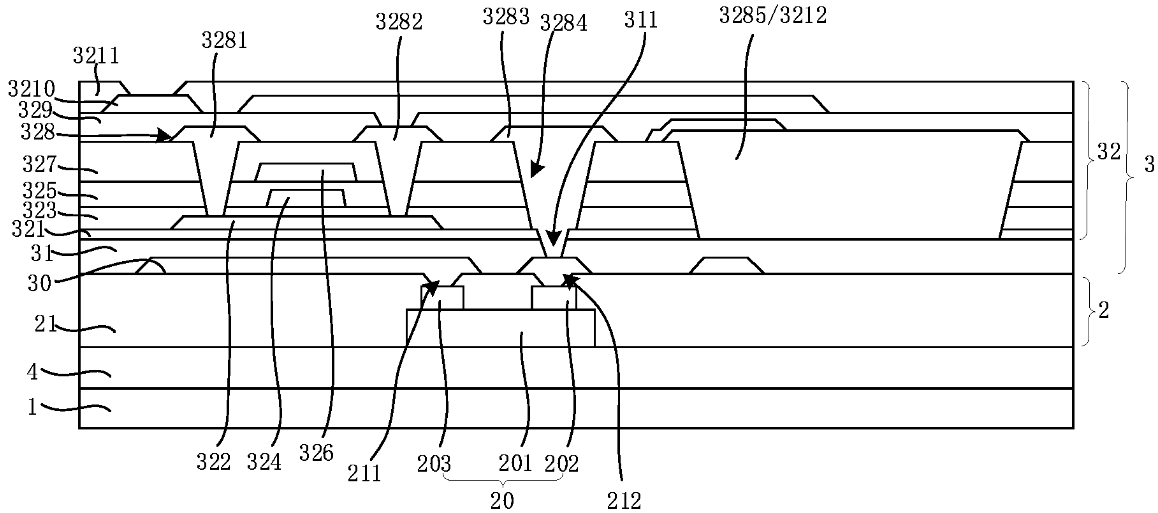
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FIG. 5H

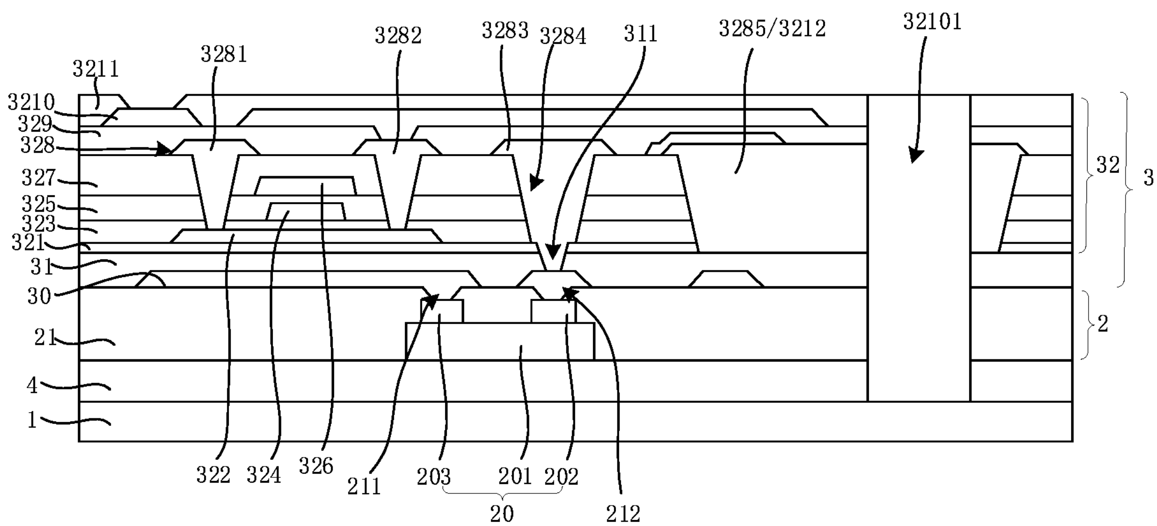
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FIG. 5I

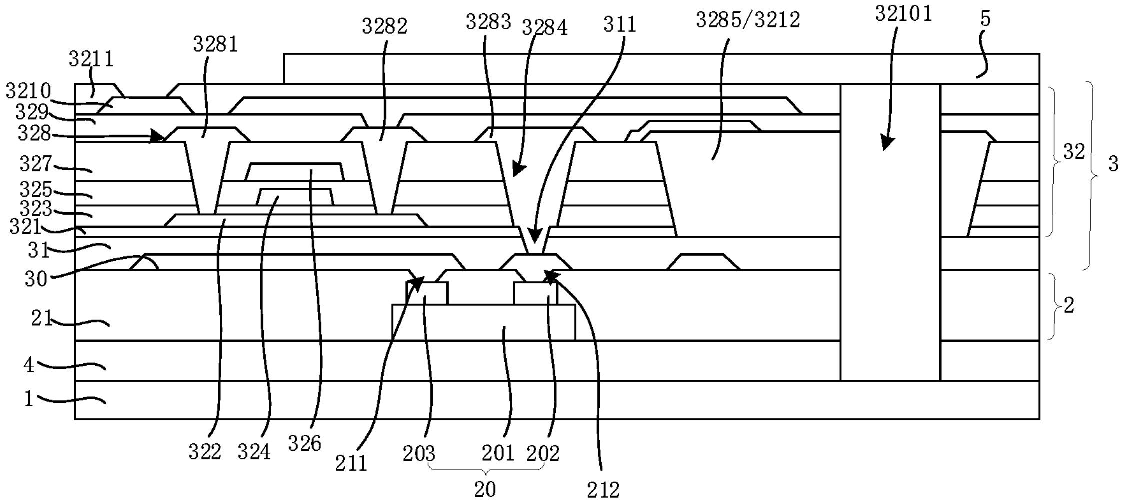
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FIG. 5J

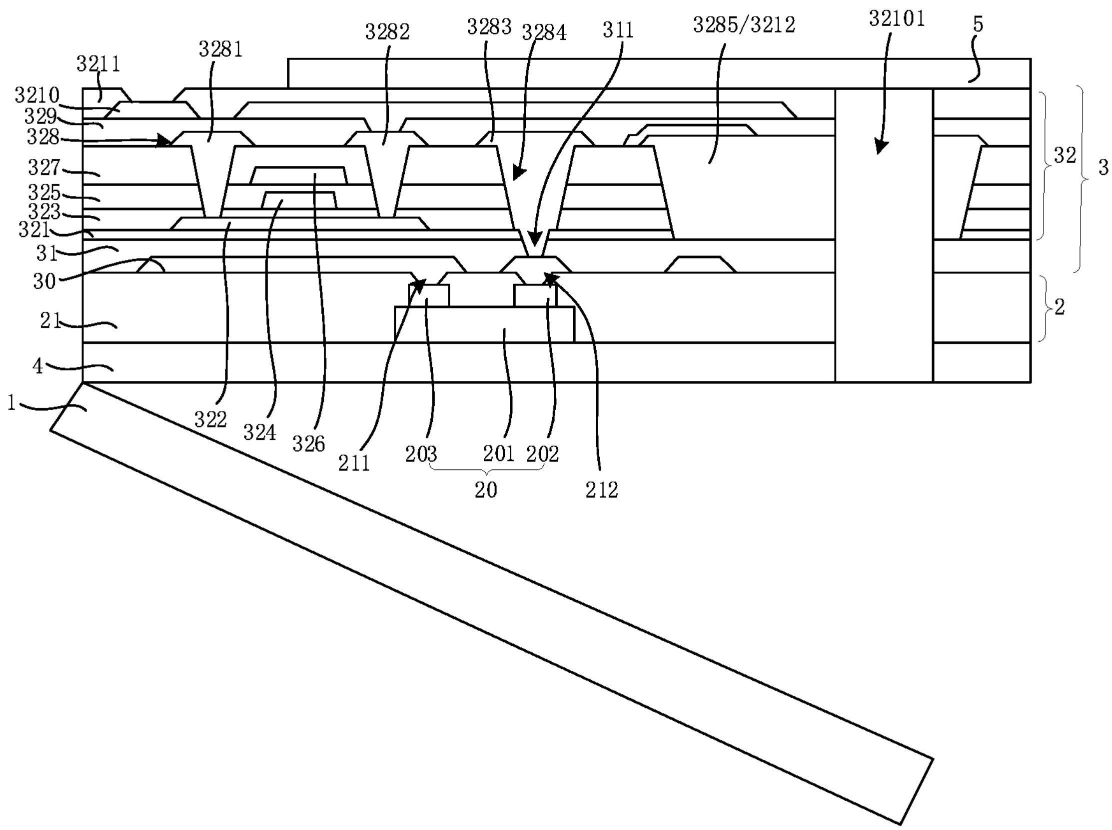
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FIG. 5K

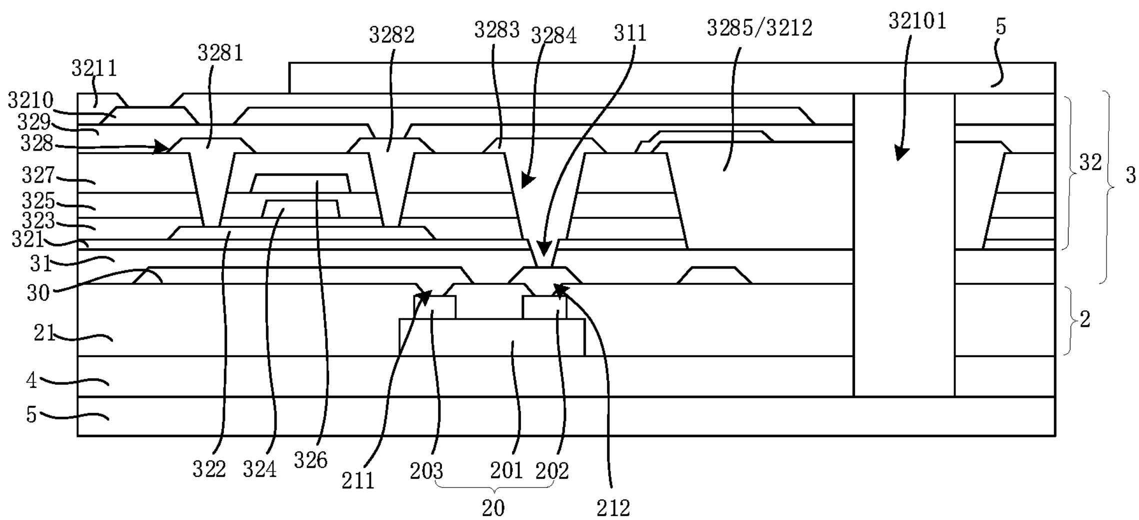
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FIG. 5L

# DISPLAY PANEL AND METHOD FOR FABRICATING THE SAME

## BACKGROUND

### Technical Field

This application relates to the technical field of display, in particular to a display panel and a method for fabricating the same.

### Related Art

Existing micro-light emitting diode (micro-LED) display panels usually adopt the following fabrication process: firstly, fabricating a driving base substrate; then, transferring LED chips on a wafer to the driving base substrate through a mass transfer process; and then, performing a module process. The mass transfer process usually used is an anisotropic conductive adhesive binding process or a metal binding process, the process is relatively more difficult, the bonding strength between the LED chip and the driving base substrate is low, and under the action of external force, the bonding between the LED chip and the driving base substrate is easily destroyed, resulting in poor stability and low yield of LED chips.

## SUMMARY

### Technical Problem

The embodiment of this application provides a display panel and a method for fabricating the same, so as to solve the technical problems of poor stability and low yield caused by the low bonding strength between the LED chip and the driving base substrate in the existing display panel and method for fabricating the same.

### Technical Solution

To resolve the foregoing problem, the technical solutions provided in this application are as follows:

The present application provides a display panel, comprising:

a base substrate, the base substrate being a glass substrate or a transparent polyimide substrate;

a micro-light emitting device layer, disposed on one side of the base substrate, a light output side of the micro-light emitting device layer facing the base substrate; and

a thin film transistor array layer, formed on the side of the micro-light emitting device layer facing away from the light output side, wherein

the micro-light emitting device layer comprises a plurality of LED chips, and the thin film transistor array layer is electrically connected to the LED chips to drive the LED chips to emit light.

According to the display panel provided by this application, the LED chip comprises a light emitting module, and a first electrode and a second electrode disposed on the side of the light emitting module away from the base substrate;

the micro-light emitting device layer further comprises a first insulating layer, the first insulating layer covers the base substrate and the LED chips, the thin film transistor array layer is electrically connected to the first electrode through a first via passing through the first insulating layer, and the thin film transistor array layer is electrically connected to the second electrode through a second via passing through the first insulating layer.



According to the display panel provided by this application, the thin film transistor array layer comprises:

a metal wiring layer, located on the side of the first insulating layer away from the base substrate, the metal wiring layer being electrically connected to the first electrode through the first via, and the metal wiring layer being electrically connected to the second electrode through the second via;

a second insulating layer, covering the first insulating layer and the metal wiring layer; and

a driving function layer, located on the side of the second insulating layer away from the base substrate, the driving function layer being electrically connected to the metal wiring layer through a third via passing through the second insulating layer.

According to the display panel provided by this application, the driving function layer comprises:

a third insulating layer, located on the side of the second insulating layer away from the base substrate;

a semiconductor layer, located on the side of the third insulating layer away from the base substrate;

a first gate insulating layer, covering the third insulating layer and the semiconductor layer;

a first gate layer, located on the side of the first gate insulating layer away from the base substrate;

a second gate insulating layer, covering the first gate layer and the first gate insulating layer;

a second gate layer, located on the side of the first gate insulating layer away from the base substrate;

an interlayer dielectric layer, covering the second gate layer and the second gate insulating layer;

a first source and drain metal layer, located on the side of the interlayer dielectric layer away from the base substrate, the first source and drain metal layer comprising a source, a drain and a signal wiring, and the signal wiring being electrically connected to the metal wiring layer through a fourth via passing through the interlayer dielectric layer, the second gate insulating layer, the first gate insulating layer and the third insulating layer, the fourth via being communicated to the third via; and

a first planarization layer, covering the interlayer dielectric layer and the first source and drain metal layer.

According to the display panel provided by this application, the display panel comprises a plurality of island regions and a plurality of hinge regions separated from each other, wherein each hinge region is located between adjacent two island regions for connecting the adjacent two island regions, and each LED chip is located in the corresponding island region; and

a fifth via is disposed in the hinge region, the fifth via passes through the driving function layer, and an organic filling layer is disposed in the fifth via.

According to the display panel provided by this application, a sixth via is disposed in the hinge region and the sixth via passes through the thin film transistor array layer and the micro-light emitting device layer.

According to the display panel provided by this application, a connection wiring part is disposed in the hinge region and the connection wiring part extends from the island region to the hinge region through the first source and drain metal layer.

According to the display panel provided by this application, the display panel further comprises a stretchable board, the stretchable board is disposed on the side of the thin film transistor array layer away from the base substrate, and/or the stretchable board is disposed on the side of the micro-light emitting device layer away from the thin film transistor array layer.

According to the display panel provided by this application, a material of the stretchable board comprises polydimethylsiloxane (PDMS).

The present application provides a display panel, comprising:

a base substrate;

a micro-light emitting device layer, disposed on one side of the base substrate, a light output side of the micro-light emitting device layer facing the base substrate; and

a thin film transistor array layer, formed on the light output side of the micro-light emitting device layer, wherein the micro-light emitting device layer comprises a plurality of LED chips, and

the thin film transistor array layer is electrically connected to the LED chips to drive the LED chips to emit light.

According to the display panel provided by this application, the LED chip comprises a light emitting module, and a first electrode and a second electrode disposed on the side of the light emitting module away from the base substrate;

the micro-light emitting device layer further comprises a first insulating layer,

the first insulating layer covers the base substrate and the LED chips, the thin film transistor array layer is electrically connected to the first electrode through a first via passing through the first insulating layer, and the thin film transistor array layer is electrically connected to the second electrode through a second via passing through the first insulating layer.

According to the display panel provided by this application, the thin film

transistor array layer comprises:

a metal wiring layer, located on the side of the first insulating layer away from the base substrate, the metal wiring layer being electrically connected to the first electrode through the first via, and the metal wiring layer being electrically connected to the second electrode through the second via;

a second insulating layer, covering the first insulating layer and the metal wiring layer; and

a driving function layer, located on the side of the second insulating layer away from the base substrate, the driving function layer being electrically connected to the metal wiring layer through a third via passing through the second insulating layer.

According to the display panel provided by this application, the driving function layer comprises:

a third insulating layer, located on the side of the second insulating layer away from the base substrate;

a semiconductor layer, located on the side of the third insulating layer away from the base substrate;

a first gate insulating layer, covering the third insulating layer and the semiconductor layer;

a first gate layer, located on the side of the first gate insulating layer away from the base substrate;

a second gate insulating layer, covering the first gate layer and the first gate insulating layer;

a second gate layer, located on the side of the first gate insulating layer away from the base substrate;

an interlayer dielectric layer, covering the second gate layer and the second gate insulating layer;

a first source and drain metal layer, located on the side of the interlayer dielectric layer away from the base substrate, the first source and drain metal layer comprising a source, a drain and a signal wiring, and the signal wiring being electrically connected to the metal wiring layer through a fourth via passing through the interlayer dielectric layer, the second gate insulating layer, the first gate insulating layer and the third insulating layer, the fourth via being communicated to the third via; and

a first planarization layer, covering the interlayer dielectric layer and the first source and drain metal layer.

According to the display panel provided by this application, the display panel comprises a plurality of island regions and a plurality of hinge regions separated from each other, wherein each hinge region is located between adjacent two island regions for connecting the adjacent two island regions, and each LED chip is located in the corresponding island region; and

a fifth via is disposed in the hinge region, the fifth via passes through the driving function layer, and an organic filling layer is disposed in the fifth via.

According to the display panel provided by this application, a sixth via is disposed in the hinge region and the sixth via passes through the thin film transistor array layer and the micro-light emitting device layer.

According to the display panel provided by this application, a connection wiring part is disposed in the hinge region and the connection wiring part extends from the island region to the hinge region through the first source and drain metal layer.

According to the display panel provided by this application, the display panel further comprises a stretchable board, the stretchable board is disposed on the side of the thin film transistor array layer away from the base substrate, and/or the stretchable board is disposed on the side of the micro-light emitting device layer away from the thin film transistor array layer.

According to the display panel provided by this application, a material of the stretchable board comprises polydimethylsiloxane (PDMS).

This application provides a method for fabricating a display panel, comprising the following steps:

providing a base substrate;

mass-transferring a plurality of LED chips to one side of the base substrate to form a micro-light emitting device layer, wherein a light output side of the micro-light emitting device layer faces the base substrate; and

forming a thin film transistor array layer on the side of the micro-light emitting device layer facing away from the light output side, and electrically connecting the thin film transistor array layer to the LED chips.

According to the method for fabricating the display panel provided by this application, after the step of mass-transferring a plurality of LED chips to one side of the base substrate, the method further comprises the following step:

detecting the plurality of LED chips, determining whether there is a faulty LED chip, and if so, repairing the faulty LED chip.

## Beneficial Effects

This application has the following beneficial effects: by forming the thin film transistor array layer on the side of the micro-light emitting device layer facing away from the light output side after mass-transferring the LED chips in the micro-light emitting device layer to the base substrate, compared with the related art, since the thin film transistor array layer is directly formed on the LED chips after mass transfer of the LED chips, the LED chips do not need to be bound and connected to the thin film transistor array layer through an anisotropic conductive adhesive binding process or a metal binding process, thus making the LED chips be firmly bonded to the thin film transistor array layer, which is beneficial to improving the stability and yield of the LED chips, and reducing the process difficulty.

## BRIEF DESCRIPTION OF THE DRAWINGS

To describe the technical solutions in the embodiments of this application more clearly, the following briefly describes the accompanying drawings required for describing the embodiments. Apparently, the accompanying drawings in the following description show merely some embodiments of this application, and a person skilled in the art may still derive other drawings from these accompanying drawings without creative efforts.

FIG. **1** illustrates a schematic sectional structural diagram of a display panel provided by an embodiment of this application.

FIG. **2** illustrates a schematic diagram of connection between an island region and a hinge region of the display panel in FIG. **1**.

FIG. **3** illustrates a schematic sectional structural diagram of another display panel provided by an embodiment of this application.

FIG. **4** illustrates a flowchart of a method for fabricating a display panel provided by an embodiment of this application.

FIG. **5**A to FIG. **5**L illustrate schematic structural diagrams of fabrication of a display panel provided by an embodiment of this application.

## DESCRIPTION OF SYMBOLS

**1**—base substrate; **2**—micro-light emitting device layer; **3**—thin film transistor array layer; **4**—fourth insulating layer; **5**—stretchable board;

**20**—LED chip; **21**—first insulating layer; **211**—first via; **212**—second via; **201**—light emitting module; **202**—first electrode; **203**—second electrode;

**30**—metal wiring layer; **31**—second insulating layer; **311**—third via; **32**—driving function layer; **321**—third insulating layer; **322**—semiconductor layer; **323**—first gate insulating layer; **324**—first gate layer; **325**—second gate insulating layer; **326**—second gate layer; **327**—interlayer dielectric layer; **328**—first source and drain metal layer; **3281**—source; **3282**—drain; **3283**—signal wiring; **3284**—fourth via; **3285**—fifth via; **329**—first planarization layer; **3210**—second source and drain metal layer; **32101**—sixth via; **3211**—second planarization layer; **3212**—organic filling layer;

**100*a***—island region; **100*b***—hinge region; and **101**—connection wiring part.

## DETAILED DESCRIPTION

The technical solutions of embodiments of the present application are clearly and completely described below with reference to the accompanying drawings in embodiments of the present application. Apparently, the described embodiments are merely a part rather than all of the embodiments of the present application. All other embodiments obtained by a person skilled in the art based on the embodiments of the present application without creative efforts shall fall within the protection scope of the present application. It should be understood that the specific implementations described herein are merely for illustrating and explaining the present application and are not intended to limit the present application. In the present application, unless otherwise specified, orientation terms such as "up" and "down" generally refer to the up and down of a device in an actual use or a working state, and specifically, refer to graphical directions of the accompanying drawings; and "inside" and "outside" refer to the inside and outside relative to the contour of the device.

Referring to FIG. **1**. FIG. **1** is a schematic sectional structural diagram of a display panel provided by an embodiment of this application. The embodiment of this application provides a display panel. The display panel includes a base substrate **1**, a micro-light emitting device layer **2** and a thin film transistor array layer **3**. The micro-light emitting device layer **2** is disposed on one side of the base substrate **1**. A light output side of the micro-light emitting device layer **2** faces the base substrate **1**. The thin film transistor array layer **3** is formed on the side of the micro-light emitting device layer **2** facing away from the light output side. The micro-light emitting device layer **2** includes a plurality of LED chips **20**. The thin film transistor array layer **3** is electrically connected to the LED chips **20** to drive the LED chips **20** to emit light.

It can be understood that this application designs a new display panel architecture. The micro-light emitting device layer **2** and the thin film transistor array layer **3** are successively disposed on the side of the base substrate **1** facing away from the light output side. Compared with the related art, in which the thin film transistor array layer **3** and the micro-light emitting device layer **2** are successively disposed on the light output side of the base substrate **1**, the thin film transistor array layer **3** is located on the side of the micro-light emitting device layer **2** facing away from the base substrate **1**, in this application, before forming the thin film transistor array layer **3**, the LED chips **20** can be mass-transferred to the side of the base substrate **1** facing away from the light output side first, and then the thin film transistor array layer **3** is directly formed on the side of the micro-light emitting device layer **2** away from the base substrate **1**. Compared with the related art, since the thin film transistor array layer **3** is directly formed on the micro-light emitting device layer **2**, the LED chips **20** do not need to be bound and connected to the thin film transistor array layer **3** through an anisotropic conductive adhesive binding process or a metal binding process during mass transfer, thus making the LED chips **20** be firmly bonded to the thin film transistor array layer **3**, which is beneficial to improving the stability and yield of the LED chips **20**, and reducing the process difficulty.

Optionally, the base substrate **1** is a transparent substrate, and the base substrate **1** is a rigid base substrate or a flexible base substrate. For example, the base substrate **1** may be a glass substrate or a transparent polyimide substrate.

Specifically, the LED chip **20** includes a light emitting module **201**, and a first electrode **202** and a second electrode **203** disposed on the side of the light emitting module **201** away from the base substrate **1**. The first electrode **202** may be one of a P electrode and an N electrode, and the second electrode **203** may be the other of a P electrode and an N electrode. In this embodiment of this application, the first electrode **202** is a P electrode, and the second electrode **203** is an N electrode.

Specifically, the micro-light emitting device layer **2** further includes a first insulating layer **21**. The first insulating layer **21** covers the base substrate **1** and the LED chip **20**. The thin film transistor array layer **3** is electrically connected to the first electrode **202** through a first via **211** passing through the first insulating layer **21**. The thin film transistor array layer **3** includes a plurality of thin film transistors distributed in an array. Each thin film transistor is electrically connected to the second electrode **203** through a second via **212** passing through the first insulating layer **21** to drive the corresponding light emitting module **201** to emit light.

While planarizing the micro-light emitting device layer **2**, the first insulating layer **21** can package the LED chip **20** without adopting an additional packaging process, thus effectively reducing the production cost.

Optionally, the first insulating layer **21** may be single-layer or multilayer. The material of the first insulating layer **21** may be an inorganic material or an organic material. For example, the inorganic material includes an oxide of silicon or a nitride of silicon or a multilayer film structure, and the organic material includes polyimide.

Specifically, a thickness of the first insulating layer **21** is in a range of 1 micron to 30 microns.

Specifically, the thin film transistor array layer **3** includes a metal wiring layer a second insulating layer **31** and a driving function layer **32**. The metal wiring layer **30** is located on the side of the first insulating layer **21** away from the base substrate **1**. The metal wiring layer **30** is electrically connected to the first electrode **202** through the first via **211**. The metal wiring layer **30** is electrically connected to the second electrode **203** through the second via **212**. The second insulating layer **31** covers the first insulating layer **21** and the metal wiring layer **30** for planarizing the metal wiring layer **30**. The driving function layer **32** is located on the side of the second insulating layer **31** away from the base substrate **1**. The driving function layer **32** is electrically connected to the metal wiring layer **30** through a third via **311** passing through the second insulating layer **31**.

Specifically, the metal wiring layer **30** includes an anode wiring and a low potential power signal line disposed at an interval. The anode wiring is electrically connected to the first electrode **202** through the first via **211**. The low potential power signal line is electrically connected to the second electrode **203** through the second via **212**.

Specifically, the driving function layer **32** includes a third insulating layer **321**, a semiconductor layer **322**, a first gate insulating layer **323**, a first gate layer **324**, a second gate insulating layer **325**, a second gate layer **326**, an interlayer dielectric layer **327**, a first source and drain metal layer **328**, and a first planarization layer **329**. The third insulating layer **321** is located on the side of the second insulating layer **31** away from the base substrate **1**. A material of the third insulating layer **321** may be an inorganic material, for preventing external water vapor from invading the driving function layer **32** and thus preventing the driving function layer **32** from being disturbed by water vapor. The semiconductor layer **322** is located on the side of the third insulating layer **321** away from the base substrate **1**. A material of the semiconductor layer **322** includes at least one of indium gallium zinc oxide (IGZO), indium gallium titanium oxide (IZTO) and indium gallium zinc titanium oxide (IGZTO). The first gate insulating layer **323** covers the third insulating layer **321** and the semiconductor layer **322**. The first gate layer **324** is located on the side of the first gate insulating layer **323** away from the base substrate **1**. The second gate insulating layer **325** covers the first gate layer **324** and the first gate insulating layer **323**. The second gate layer **326** is located on the side of the first gate insulating layer **323** away from the base substrate **1**. The interlayer dielectric layer **327** covers the second gate layer **326** and the second gate insulating layer **325**. The first source and drain metal layer **328** is located on the side of the interlayer dielectric layer **327** away from the base substrate **1**. The first source and drain metal layer **328** includes a source **3281**, a drain **3282** and a signal wiring **3283**. The signal wiring **3283** is electrically connected to the metal wiring layer **30** through a fourth via **3284** passing through the interlayer dielectric layer **327**, the second gate insulating layer **325**, the first gate insulating layer **323** and the third insulating layer **321**. The fourth via **3284** is communicated with the third via **311**. The first planarization layer **329** covers the interlayer dielectric layer **327** and the first source and drain metal layer **328**.

Specifically, the semiconductor layer **322** includes a channel region, and a source region and a drain region located on two sides of the channel region. The source **3281** is electrically connected to the source region through a source contact of the first gate insulating layer **323** passing through the interlayer dielectric layer **327** and the second gate insulating layer **325**. The drain **3282** is electrically connected to the drain region through a drain contact of the first gate insulating layer **323** passing through the interlayer dielectric layer **327** and the second gate insulating layer **325**.

Specifically, the signal wiring **3283** includes functional signal lines such as a data line, a scanning line, a driving voltage line and a voltage transmission line, which are used to transmit data signals, scanning signals, driving voltage signals and voltage transmission signals respectively. For example, when the signal wiring **3283** is a data line, it is used to transmit a data signal to the LED chip **20** through the metal wiring layer **30**.

It should be noted that the structure of a single thin film transistor in the driving function layer **32** is not limited to a double-gate structure provided by an embodiment of this application. Those skilled in the art can also choose other structural forms, such as a single-gate structure, which will not be repetitively described here.

Further, in order to reduce voltage drop, the driving function layer **32** may also adopt a double-layer source and drain metal layer design. Specifically, the driving function layer **32** further includes a second source and drain metal layer **3210** and a second planarization layer **3211**. The second source and drain metal layer **3210** is located on the side of the first planarization layer **329** away from the base substrate **1**. The second source and drain metal layer **3210** is electrically connected to the drain **3282** through a via passing through the first planarization layer **329**. The second planarization layer **3211** covers the first planarization layer **329** and the second source and drain metal layer **3210**.

In this embodiment of this application, a fourth insulating layer **4** is further disposed between the base substrate **1** and the micro-light emitting device layer **2**, for preventing the failure of the light emitting module **201** caused by external water vapor invading the micro-light emitting device layer **2**. Of course, in other embodiments, the fourth insulating layer **4** may not be disposed.

Further, jointly referring to FIG. **1** and FIG. **2**. FIG. **2** is a schematic diagram of connection between an island region and a hinge region of the display panel in FIG. **1**. The display panel may be a display panel with stretchability characteristics. Specifically, the display panel includes a plurality of island regions **100***a* and a plurality of hinge regions **100***b* separated from each other. The hinge region **100***b* is located between adjacent two island regions **100***a* for connecting the adjacent two island regions **100***a*. The plurality of LED chips is located in the corresponding island regions **100***a* respectively.

It can be understood that the island region **100***a* is used to realize display, and the hinge region **100***b* is used to realize ductile deformation. When the display panel is folded, compressed or stretched under external force, the hinge region **100***b* generates ductile deformation in different directions in a plane under the action of tensile force in each direction, a gap between the adjacent island regions **100***a* will change accordingly, and the island regions **100***a* will rotate in a follow-up manner, so as to realize the stretchability characteristics. When the external force disappears, the hinge region **100***b* will automatically return to an original state, and the display panel will return to an original state.

Further, a fifth via **3285** is disposed in the hinge region **100***b*, and the fifth via **3285** passes through the driving function layer **32**. In this embodiment of this application, the fifth via **3285** passes through the interlayer dielectric layer **327**, the second gate insulating layer **325**, the first gate insulating layer **323** and the third insulating layer **321**. An organic filling layer **3212** is disposed in the fifth via **3285**. The organic filling layer **3212** functions to improve flexibility, reduce bending stress and avoid the fracture of the signal line **3283** when the hinge region **100***b* is bent.

Further, a sixth via **32101** is further disposed in the hinge region **100***b*, and the sixth via **32101** passes through the thin film transistor array layer **3** and the micro-light emitting device layer **2**, so as to make the hinge region **100***b* stretchable. In this embodiment of this application, the sixth via **32101** exposes a surface of the side of the base substrate **1** close to the micro-light emitting device layer **2**, and the sixth via **32101** passes through the interlayer dielectric layer **327**, the second gate insulating layer **325**, the first gate insulating layer **323**, the third insulating layer **321**, the second insulating layer **31** and the first insulating layer **21**.

Further, a connection wiring part **101** is disposed in the hinge region **100***b* and the connection wiring part **101** extends from the island region **100***a* to the hinge region **100***b* through the first source and drain metal layer **328**. When the driving function layer **32** can also adopt a double-layer source and drain metal layer design, the connection wiring part **101** extends from the island region **100***a* to the hinge region **100***b* through the first source and drain metal layer **328** and the second source and drain metal layer **3210**.

Further, the display panel further includes a stretchable board **5**, the stretchable board **5** is disposed on the side of the thin film transistor array layer **3** away from the base substrate **1**, and/or the stretchable board **5** is disposed on the side of the micro-light emitting device layer **2** close to the base substrate **1**. It can be understood that the stretchable board **5** has stretchability, thus further improving the overall stretchability of the display panel.

Further, referring to FIG. **3**. FIG. **3** is a schematic sectional structural diagram of another display panel provided by an embodiment of this application. The stretchable board **5** can also play a role of protecting the display panel. When the stretchable board **5** is disposed on the side of the micro-light emitting device layer **2** away from the thin film transistor array layer **3**, the stretchable board **5** can replace the base substrate **1**. That is, the base substrate **1** can be peeled off after the stretchable board **5** is attached to the display panel, thus helping to reduce a thickness of the display panel and further improving the flexibility of the display panel.

Optionally, a material of the stretchable board **5** includes a material with stretchability such as polydimethylsiloxane (PDMS).

Referring to FIG. **4** and FIG. **5**A to FIG. **5**L. FIG. **4** is a flowchart of a method for fabricating a display panel provided by an embodiment of this application. FIG. **5**A to FIG. **5**L are schematic structural diagrams of fabrication of a display panel provided by an embodiment of this application. The embodiment of this application further provides a method for fabricating a display panel. The method includes the following steps:

S**10**: Provide a base substrate **1**.

S**20**: Mass-transfer a plurality of LED chips **20** to one side of the base substrate **1** to form a micro-light emitting device layer **2**, wherein a light output side of the micro-light emitting device layer **2** faces the base substrate **1**.

S**30**: Form a thin film transistor array layer **3** on the side of the micro-light emitting device layer **2** facing away from the light output side, and electrically connect the thin film transistor array layer **3** to the LED chips **20**.

Specifically, referring to FIG. **5**A, in step S**10**, a base substrate **1** is provided, and a fourth insulating layer **4** is deposited on one side of the base substrate **1**.

Specifically, referring to FIG. **5**B, in step S**20**, the LED chip **20** includes a light emitting module **201**, and a first electrode **202** and a second electrode **203** disposed on the side of the light emitting module **201** away from the base substrate **1**. In this embodiment of this application, the first electrode **202** is a P electrode and the second electrode **203** is an N electrode.

Further, after the step of mass-transferring a plurality of LED chips **20** to one side of the base substrate **1**, the method further includes the following step:

S**201**: Detect a plurality of the LED chips **20**, determine whether there is a faulty LED chip, and if so, repair the faulty LED chip.

It can be understood that, in this application, the LED chips **20** are detected before forming the thin film transistor array layer **3**. Compared with the related art, it can avoid the situation that the whole display panel is discarded as useless since the LED chips are detected after mass-transferring the LED chips **20** to the thin film transistor array layer **3** and a faulty LED chip cannot be repaired even if detected, thus greatly reducing the production cost.

Specifically, after step S**201**, the method further includes the following steps:

S**202**: Form a first insulating layer **21** covering the base substrate **1** and the LED chips **20**.

S**203**: Form a first via **211** and a second via **212** passing through the first insulating layer **21**.

Specifically, referring to FIG. **5**C, a yellow light process is adopted for the first insulating layer **21** to form the first via **211** and the second via **212**. The first via **211** exposes a surface of the side of the first electrode away from the light emitting module. The second via **212** exposes a surface of the side of the second electrode **203** away from the light emitting module **201**.

Specifically, step S**30** includes the following steps:

S**301**: Form a metal wiring layer **30** on the side of the first insulating layer **21** away from the base substrate **1**.

Specifically, referring to FIG. **5**D, the metal wiring layer **30** may be formed on the side of the first insulating layer **21** away from the base substrate **1** by adopting a yellow light process. The metal wiring layer **30** includes an anode wiring and a low potential power signal line disposed at an interval. The anode wiring is electrically connected to the first electrode **202** through the first via **211**. The low potential power signal line is electrically connected to the second electrode **203** through the second via **212**.

S**302**: Form a second insulating layer **31** covering the first insulating layer **21** and the metal wiring layer **30**.

Specifically, referring to FIG. **5**E, the second insulating layer **31** can be formed through a deposition process, and a material of the second insulating layer **31** may be an organic material for planarizing the metal wiring layer **30**; and then, the second insulating layer **31** can be patterned through a yellow light process to form a third via **311**.

S**303**: Form a driving function layer on the side of the second insulating layer **31** away from the base substrate **1**.

Specifically, step S**303** includes the following steps:

S**3031**: Form a third insulating layer **321** on the side of the second insulating layer **31** away from the base substrate **1**.

S**3032**: Form a semiconductor layer **322** on the side of the third insulating layer **321** away from the base substrate **1**.

S**3033**: Form a first gate insulating layer **323** covering the third insulating layer **321** and the semiconductor layer **322**.

S**3034**: Form a first gate layer **324** on the side of the first gate insulating layer **323** away from the base substrate **1**.

S**3035**: Form a second gate insulating layer **325** covering the first gate layer **324** and the first gate insulating layer **323**.

S**3036**: Form a second gate layer **326** on the side of the first gate insulating layer **323** away from the base substrate **1**;

S**3037**: Form an interlayer dielectric layer **327** covering the second gate layer **326** and the second gate insulating layer **325**.

S**3038**: Form a first source and drain metal layer **328** on the side of the interlayer dielectric layer **327** away from the base substrate **1**, and pattern the first source and drain metal layer **328** to form a source **3281**, a drain **3282** and a signal wiring **3283**.

S**3039**: Form a first planarization layer **329** covering the interlayer dielectric layer **327** and the first source and drain metal layer **328**.

Specifically, referring to FIG. **5**F, after step S**3037**, a source contact, a drain contact, a fourth via **3284** and a fifth via **3285** can be formed through a yellow light process. The source **3281** is electrically connected to the source region of the semiconductor layer **322** through the source contact. The drain **3282** is electrically connected to the drain region of the semiconductor layer **322** through the drain contact. The fourth via **3284** passes through the interlayer dielectric layer **327**, the second gate insulating layer **325**, the first gate insulating layer **323** and the third insulating layer **321**. The fourth via **3284** is communicated with the third via **311**. The fifth via **3285** passes through the interlayer dielectric layer **327**, the second gate insulating layer **325**, the first gate insulating layer **323** and the third insulating layer **321**.

Referring to FIG. **5**G, after the fifth via **3285** is formed, an organic filling layer **3212** is filled into the fifth via **3285**, and a material of the organic filling layer **3212** is an organic material.

Further, referring to FIG. **5**H, after step S**3039**, the method further includes: forming a second source and drain metal layer **3210** on the side of the first planarization layer **329** away from the base substrate **1**, wherein the second source and drain metal layer **3210** is electrically connected to the drain **3282** through a via passing through the first planarization layer **329**; and forming a second planarization layer **3211** covering the first planarization layer **329** and the second source and drain metal layer **3210**.

Further, after step S**3039**, a sixth via **32101** passing through the interlayer dielectric layer **327**, the second gate insulating layer **325**, the first gate insulating layer **323**, the third insulating layer **321**, the second insulating layer **31** and the first insulating layer **21** can be formed in the hinge region **100***b* through a yellow light process.

Further, after step S**30**, the method for fabricating the display panel further includes the following step:

S**40**: Attach a stretchable board **5** to the side of the thin film transistor array layer **3** away from the base substrate **1** and/or the side of the micro-light emitting device layer **2** away from the thin film transistor array layer **3**.

Specifically, in one embodiment, referring to FIG. **5**I, the stretchable board **5** is attached to the side of the thin film transistor array layer **3** away from the base substrate **1**.

In one embodiment, referring to FIG. **5**J, the base substrate **1** is peeled off; referring to FIG. **5**K, the stretchable board **5** is attached to the side of the micro-light emitting device layer **2** away from the thin film transistor array layer **3**.

This application has the following beneficial effects: by forming the thin film transistor array layer on the side of the micro-light emitting device layer facing away from the light output side after mass-transferring the LED chips in the micro-light emitting device layer to the base substrate, compared with the related art, since the thin film transistor array layer is directly formed on the LED chips after mass transfer of the LED chips, the LED chips do not need to be bound and connected to the thin film transistor array layer through an anisotropic conductive adhesive binding process or a metal binding process, thus making the LED chips be firmly bonded to the thin film transistor array layer, which is beneficial to improving the stability and yield of the LED chips, and reducing the process difficulty.

In summary, the present application has been disclosed through preferred embodiments; however, the preferred embodiments are not intended to limit the present application, and a person of ordinary skill in the art can make various modifications and improvements without departing from the spirit and scope of the present application; therefore, the protection scope of the present application should be subject to the scope defined by the claims.

What is claimed is:

**1**. A display panel, comprising:

a base substrate, wherein the base substrate is a glass substrate or a transparent polyimide substrate;

a micro-light emitting device layer, disposed on one side of the base substrate, wherein a light output side of the micro-light emitting device layer faces the base substrate; and

a thin film transistor array layer, disposed on the side of the micro-light emitting device layer facing away from the light output side;

wherein the micro-light emitting device layer comprises a plurality of light emitting diode (LED) chips, and the thin film transistor array layer is electrically connected to the LED chips to drive the LED chips to emit light;

wherein the display panel further comprises a plurality of island regions and a plurality of hinge regions separated from each other, each hinge region is located between adjacent two island regions for connecting the adjacent two island regions, and each LED chip is located in the corresponding island region; a sixth via is disposed in the hinge region, and the sixth via passes through the thin film transistor array layer and the micro-light emitting device layer;

wherein the LED chip comprises a light emitting module, and a first electrode and a second electrode disposed on the side of the light emitting module away from the base substrate;

wherein the micro-light emitting device layer further comprises a first insulating layer, the first insulating layer covers the base substrate and the LED chips, the thin film transistor array layer is electrically connected to the first electrode through a first via passing through the first insulating layer, and the thin film transistor array layer is electrically connected to the second electrode through a second via passing through the first insulating layer;

wherein the thin film transistor array layer comprises:

a metal wiring layer, located on the side of the first insulating layer away from the base substrate, wherein the metal wiring layer is electrically connected to the first electrode through the first via, and the metal wiring layer is electrically connected to the second electrode through the second via;

a second insulating layer, covering the first insulating layer and the metal wiring layer; and

a driving function layer, located on the side of the second insulating layer away from the base substrate, wherein the driving function layer is electrically connected to the metal wiring layer through a third via passing through the second insulating layer;

wherein the driving function layer comprises:

a third insulating layer, located on the side of the second insulating layer away from the base substrate;

a semiconductor layer, located on the side of the third insulating layer away from the base substrate;

a first gate insulating layer, covering the third insulating layer and the semiconductor layer;

a first gate layer, located on the side of the first gate insulating layer away from the base substrate;

a second gate insulating layer, covering the first gate layer and the first gate insulating layer;

a second gate layer, located on the side of the first gate insulating layer away from the base substrate;

an interlayer dielectric layer, covering the second gate layer and the second gate insulating layer;

a first source and drain metal layer, located on the side of the interlayer dielectric layer away from the base substrate, wherein the first source and drain metal layer comprise a source, a drain and a signal wiring, and the signal wiring is electrically connected to the metal wiring layer through a fourth via passing through the interlayer dielectric layer, the second gate insulating layer, the first gate insulating layer and the third insulating layer, the fourth via is communicated with the third via; and

a first planarization layer, covering the interlayer dielectric layer and the first source and drain metal layer; and

wherein the sixth via exposes a surface of the side of the base substrate close to the micro-light emitting device layer, and the sixth via passes through the interlayer dielectric layer, the second gate insulating layer, the first gate insulating layer, the third insulating layer, the second insulating layer and the first insulating layer.

**2**. The display panel according to claim **1**, wherein a connection wiring part is disposed in the hinge region and the connection wiring part extends from the island region to the hinge region through the first source and drain metal layer.

**3**. The display panel according to claim **1**, further comprising a stretchable board, wherein the stretchable board is disposed on the side of the thin film transistor array layer away from the base substrate, or the stretchable board is disposed on the side of the micro-light emitting device layer away from the thin film transistor array layer.

**4**. The display panel according to claim **3**, wherein a material of the stretchable board comprises polydimethylsiloxane (PDMS).

**5**. A display panel, comprising:

a base substrate;

a micro-light emitting device layer, disposed on one side of the base substrate, wherein a light output side of the micro-light emitting device layer faces the base substrate; and

a thin film transistor array layer, disposed on the side of the micro-light emitting device layer facing away from the light output side;

wherein the micro-light emitting device layer comprises a plurality of light emitting diode (LED) chips, and the thin film transistor array layer is electrically connected to the LED chips to drive the LED chips to emit light;

wherein the display panel further comprises a plurality of island regions and a plurality of hinge regions separated from each other, each hinge region is located between adjacent two island regions for connecting the adjacent two island regions, and each LED chip is located in the corresponding island region; a sixth via is disposed in the hinge region, and the sixth via passes through the thin film transistor array layer and the micro-light emitting device layer;

wherein the LED chip comprises a light emitting module, and a first electrode and a second electrode disposed on the side of the light emitting module away from the base substrate;

wherein the micro-light emitting device layer further comprises a first insulating layer, the first insulating layer covers the base substrate and the LED chips, the thin film transistor array layer is electrically connected to the first electrode through a first via passing through the first insulating layer, and the thin film transistor array layer is electrically connected to the second electrode through a second via passing through the first insulating layer;

wherein the thin film transistor array layer comprises:

a metal wiring layer, located on the side of the first insulating layer away from the base substrate, wherein the metal wiring layer is electrically connected to the first electrode through the first via, and the metal wiring layer is electrically connected to the second electrode through the second via;

a second insulating layer, covering the first insulating layer and the metal wiring layer; and

a driving function layer, located on the side of the second insulating layer away from the base substrate, wherein the driving function layer is electrically connected to the metal wiring layer through a third via passing through the second insulating layer;

wherein the driving function layer comprises:

a third insulating layer, located on the side of the second insulating layer away from the base substrate;

a semiconductor layer, located on the side of the third insulating layer away from the base substrate;

a first gate insulating layer, covering the third insulating layer and the semiconductor layer;

a first gate layer, located on the side of the first gate insulating layer away from the base substrate;

a second gate insulating layer, covering the first gate layer and the first gate insulating layer;

a second gate layer, located on the side of the first gate insulating layer away from the base substrate;

an interlayer dielectric layer, covering the second gate layer and the second gate insulating layer;

a first source and drain metal layer, located on the side of the interlayer dielectric layer away from the base substrate, wherein the first source and drain metal layer comprise a source, a drain and a signal wiring, and the signal wiring is electrically connected to the metal wiring layer through a fourth via passing through the interlayer dielectric layer, the second gate insulating layer, the first gate insulating layer and the third insulating layer, the fourth via is communicated with the third via; and

a first planarization layer, covering the interlayer dielectric layer and the first source and drain metal layer; and

wherein the sixth via exposes a surface of the side of the base substrate close to the micro-light emitting device layer, and the sixth via passes through the interlayer dielectric layer, the second gate insulating layer, the first gate insulating layer, the third insulating layer, the second insulating layer and the first insulating layer.

**6**. The display panel according to claim **5**, wherein a connection wiring part is disposed in the hinge region and the connection wiring part extends from the island region to the hinge region through the first source and drain metal layer.

**7**. The display panel according to claim **5**, further comprising a stretchable board, wherein the stretchable board is disposed on the side of the thin film transistor array layer away from the base substrate, or the stretchable board is disposed on the side of the micro-light emitting device layer away from the thin film transistor array layer.

**8**. The display panel according to claim **7**, wherein a material of the stretchable board comprises polydimethylsiloxane (PDMS).

**9**. A method for fabricating a display panel, comprising following steps:

providing a base substrate;

mass-transferring a plurality of light emitting diode (LED) chips to one side of the base substrate to form a micro-light emitting device layer, wherein a light output side of the micro-light emitting device layer faces the base substrate; and

forming a thin film transistor array layer on the side of the micro-light emitting device layer facing away from the light output side, and electrically connecting the thin film transistor array layer to the LED chips;

wherein the display panel further comprises a plurality of island regions and a plurality of hinge regions separated from each other, each hinge region is located between adjacent two island regions for connecting the adjacent two island regions, and each LED chip is located in the corresponding island region; a sixth via is disposed in the hinge region, and the sixth via passes through the thin film transistor array layer and the micro-light emitting device layer;

wherein the LED chip comprises a light emitting module, and a first electrode and a second electrode disposed on the side of the light emitting module away from the base substrate;



wherein the micro-light emitting device layer further comprises a first insulating layer, the first insulating layer covers the base substrate and the LED chips, the thin film transistor array layer is electrically connected to the first electrode through a first via passing through the first insulating layer, and the thin film transistor array layer is electrically connected to the second electrode through a second via passing through the first insulating layer;

wherein the thin film transistor array layer comprises:

a metal wiring layer, located on the side of the first insulating layer away from the base substrate, wherein the metal wiring layer is electrically connected to the first electrode through the first via, and the metal wiring layer is electrically connected to the second electrode through the second via;

a second insulating layer, covering the first insulating layer and the metal wiring layer; and

a driving function layer, located on the side of the second insulating layer away from the base substrate, wherein the driving function layer is electrically connected to the metal wiring layer through a third via passing through the second insulating layer;

wherein the driving function layer comprises:

a third insulating layer, located on the side of the second insulating layer away from the base substrate;

a semiconductor layer, located on the side of the third insulating layer away from the base substrate;

a first gate insulating layer, covering the third insulating layer and the semiconductor layer;

a first gate layer, located on the side of the first gate insulating layer away from the base substrate;

a second gate insulating layer, covering the first gate layer and the first gate insulating layer:

a second gate layer, located on the side of the first gate insulating layer away from the base substrate;

an interlayer dielectric layer, covering the second gate layer and the second gate insulating layer;

a first source and drain metal layer, located on the side of the interlayer dielectric layer away from the base substrate, wherein the first source and drain metal layer comprise a source, a drain and a signal wiring, and the signal wiring is electrically connected to the metal wiring layer through a fourth via passing through the interlayer dielectric layer, the second gate insulating layer, the first gate insulating layer and the third insulating layer, the fourth via is communicated with the third via; and

a first planarization layer, covering the interlayer dielectric layer and the first source and drain metal layer; and

wherein the sixth via exposes a surface of the side of the base substrate close to the micro-light emitting device layer, and the sixth via passes through the interlayer dielectric layer, the second gate insulating layer, the first gate insulating layer, the third insulating layer, the second insulating layer and the first insulating layer.

**10**. The method for fabricating the display panel according to claim **9**, wherein after the step of mass-transferring a plurality of LED chips to one side of the base substrate, the method further comprises following steps:

detecting the plurality of LED chips, determining whether there is a faulty LED chip, and if so, repairing the faulty LED chip.

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